HDOC Day-55

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Problem Statement

Write a C program that reads the size and elements of a variable-length integer array **arr[]**, and an integer **k** such that 0 < k < size. Pass the array to a function named **window_maximum()** using its address. Use pointers to pass the array between the calling function and **window_maximum()**.

Consider every contiguous subarray of size **k** as a sliding window. Find and display the maximum element in each window from left to right, separated by spaces. Display **-1** if **k** is invalid. Use a nested-loop approach.

C Program

```
#include <stdio.h>

void window_maximum(int *arr, int size, int k)
{
    int i, j, max;

    if (k <= 0 || k >= size)
    {
        printf("-1\n");
        return;
    }

    for (i = 0; i <= size - k; i++)
    {
        max = *(arr + i);

        for (j = i + 1; j < i + k; j++)
        {
            if (*(arr + j) > max)
                max = *(arr + j);
        }

        printf("%d", max);

        if (i < size - k)
            printf(" ");
    }

    printf("\n");
}

int main()
{
    int size, i, k;

    printf("Enter size of array: ");
    scanf("%d", &size);

    int arr[size];

    printf("Enter array elements: ");
    for (i = 0; i < size; i++)
        scanf("%d", arr + i);

    printf("Enter k: ");
    scanf("%d", &k);

    printf("Output: ");
    window_maximum(arr, size, k);
```

```
    return 0;
}
```

Explanation

The array address is passed to **window_maximum()** using a pointer. The outer loop moves the sliding window from left to right. For each window of size **k**, the inner loop compares its elements and finds the maximum value. That maximum is printed before moving to the next window. If **k** is outside the valid range, the function prints **-1**.

Sample Outputs

```
Example 1
Array: 1 3 -1 -3 5 3 6 7
k: 3
Output: 3 3 5 5 6 7

Example 2
Array: 1 2 3 4 5
k: 2
Output: 2 3 4 5

Example 3
Array: 1 2 3
k: 4
Output: -1
```

Result

Thus, the program successfully finds the maximum element in every contiguous sliding window of size k using pointers and nested loops, and displays -1 when k is invalid.